

Derivation 1:

(a) Eq of motion for individual particles:

$$m_1 \frac{d^2 \vec{r}_1}{dt^2} = \vec{F}_1^{(e)} + \vec{F}_{12}$$

$$m_2 \frac{d^2 \vec{r}_2}{dt^2} = \vec{F}_2^{(e)} + \vec{F}_{21}$$

+ add them up $\Rightarrow$

$$\Rightarrow \frac{d^2}{dt^2} (m_1 \vec{r}_1 + m_2 \vec{r}_2) = \vec{F}_1^{(e)} + \vec{F}_2^{(e)} + \vec{F}_{12} + \vec{F}_{21}$$

Center of mass position vector

$$\vec{R} = \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2}{m_1 + m_2} \Rightarrow M \vec{R} = m_1 \vec{r}_1 + m_2 \vec{r}_2$$

$M = m_1 + m_2$

$$\Rightarrow M \frac{d^2 \vec{R}}{dt^2} = \vec{F}^{(e)} + \vec{F}_{12} + \vec{F}_{21}$$

$\vec{F}^{(e)} = \vec{F}_1^{(e)} + \vec{F}_2^{(e)}$

but, if the system obeys $M \frac{d^2 \vec{R}}{dt^2} = \vec{F}^{(e)}$ $\Rightarrow \vec{F}_{12} + \vec{F}_{21} = 0$ which is the weak law of action & reaction

(b) Eq of motion for particle 1:

$$\frac{d\vec{p}_1}{dt} = \vec{F}_1^{(e)} + \vec{F}_{12} \quad \text{but} \quad \frac{d}{dt} (\vec{r} \times m\vec{v}) = \vec{v} \times m\vec{v} + \vec{r} \times \frac{d}{dt} m\vec{v}$$

$\vec{v} \times m\vec{v} = 0$

$$\text{So } \vec{r}_1 \times \frac{d\vec{p}_1}{dt} = \vec{r}_1 \times \vec{F}_1^{(e)} + \vec{r}_1 \times \vec{F}_{12}$$

$$\Rightarrow \frac{d}{dt} (\vec{r}_1 \times \vec{p}_1) = \vec{r}_1 \times \vec{F}_1^{(e)} + \vec{r}_1 \times \vec{F}_{12} \Rightarrow \frac{d\vec{L}_1}{dt} = \vec{N}_1^{(e)} + \vec{r}_1 \times \vec{F}_{12} \quad (*)$$

Similarly, for the 2nd particle

$$\frac{d\vec{L}_2}{dt} = \vec{N}_2^{(e)} + \vec{r}_2 \times \vec{F}_{21} \quad (*)$$

add (*) to (*)

$$\Rightarrow \frac{d}{dt} (\vec{L}_1 + \vec{L}_2) = \vec{N}_1^{(e)} + \vec{N}_2^{(e)} + \vec{r}_1 \times \vec{F}_{12} + \vec{r}_2 \times \vec{F}_{21}$$

$\vec{F}_{12} = -\vec{F}_{21}$
(we showed this @ point (a))

$$\Rightarrow \frac{d}{dt} \vec{L} = \vec{N}^{(e)} + (\vec{r}_1 - \vec{r}_2) \times \vec{F}_{12}$$

but if the system obeys $\frac{d\vec{L}}{dt} = \vec{N}^{(e)}$ $\Rightarrow$

$\Rightarrow (\vec{r}_1 - \vec{r}_2) \times \vec{F}_{12} = 0$ i.e. $\vec{F}_{12}$ is along the direction that joins the two particles $(\vec{r}_1 - \vec{r}_2)$ thus the strong law of action & reaction holds.

Derivation 2

(a) for circular orbits: $r_c = \frac{\ell^2}{mk}$

for parabolic orbits: $r_p = \frac{\ell^2}{mk} \left(\frac{1}{1 + \cos \theta} \right)$

Now, at the perihelion of the parabola (i.e. for $\theta = 0$) we get

$$r_p = \frac{\ell^2}{2mk} = \frac{r_c}{2} \quad \checkmark$$

(b) for the parabola:

$$r = \frac{\ell^2}{mk} \left(\frac{1}{1 + \cos \theta} \right)$$

$$\dot{r} = \frac{\ell^2}{mk} \left[\frac{\sin \theta}{(1 + \cos \theta)^2} \right] \dot{\theta} = \dot{\theta} \frac{\ell^2 \sin \theta}{mk(1 + \cos \theta)^2}$$

The velocity @ coordinate r, θ is:

$$v^2 = \dot{r}^2 + r^2 \dot{\theta}^2 = r^2 \dot{\theta}^2 \left[\frac{\sin^2 \theta}{(1 + \cos \theta)^2} + 1 \right] = r^2 \dot{\theta}^2 \left[\frac{\sin^2 \theta + 1 + 2 \cos \theta + \cos^2 \theta}{(1 + \cos \theta)^2} \right] =$$

$$= 2r^2 \dot{\theta}^2 \left[\frac{1}{1 + \cos \theta} \right] = \frac{2mk r^3 \dot{\theta}^2}{\ell^2} = \frac{2K}{mr} \quad (*) \text{ as } \ell = m r^2 \dot{\theta}$$

for the circular trajectory, using $\frac{k}{r} = \frac{l^2}{mr^2}$ we find
 $\sqrt{v} = r^2 \dot{\theta}^2 = \frac{l^2}{mr^2} = \frac{l^2}{mr}$ (*)
 Comparing (*) to (**) we find that the speed in the parabolic trajectory is $\sqrt{2}$ times greater than its circular counterpart.

Problem 1

Positions of the masses are:
$$\begin{cases} \mathbf{r}_1 = l \cdot (\sin\theta \cos\Omega t, \sin\theta \sin\Omega t, \cos\theta) \\ \mathbf{r}_2 = (0, 0, 2l \cos\theta) \end{cases}$$

The two masses have velocities:

$$\begin{cases} \mathbf{v}_1 = l \dot{\theta} (\cos\theta \cos\Omega t, \cos\theta \sin\Omega t, -\sin\theta) + l\Omega (-\sin\theta \sin\Omega t, \sin\theta \cos\Omega t, 0) \\ \mathbf{v}_2 = -2l \dot{\theta} (0, 0, \sin\theta) \end{cases}$$

The kinetic energy is:

$$T = \frac{1}{2} m v_1^2 + \frac{1}{2} m v_2^2 = \frac{1}{2} m l^2 (\dot{\theta}^2 + \Omega^2 \sin^2\theta) + \frac{1}{2} m (4l^2 \sin^2\theta) \dot{\theta}^2$$

The potential energy is:

$$V = -mgz_1 - mgz_2 = -3mgl \cos\theta$$

Finally, the Lagrangian is:

$$L = \frac{1}{2} m l^2 (\dot{\theta}^2 + \Omega^2 \sin^2\theta) + 2m l^2 \dot{\theta}^2 \sin^2\theta + 3mgl \cos\theta$$

Then, the Lagrange equation $\frac{\partial}{\partial t} \left(\frac{\partial L}{\partial \dot{\theta}} \right) - \frac{\partial L}{\partial \theta} = 0$ becomes:

$$\begin{aligned} \frac{d}{dt} (m l^2 \dot{\theta} + 4m l^2 \dot{\theta} \sin^2\theta) - (m l^2 \Omega^2 \sin\theta \cos\theta + 4m l^2 \dot{\theta}^2 \sin\theta \cos\theta - 3mgl \sin\theta) &= 0 \\ m l^2 \ddot{\theta} + 4m l^2 \ddot{\theta} \sin^2\theta + 8m l^2 \dot{\theta}^2 \sin\theta \cos\theta - (m l^2 \Omega^2 \sin\theta \cos\theta + 4m l^2 \dot{\theta}^2 \sin\theta \cos\theta - 3mgl \sin\theta) &= 0 \\ m l^2 \ddot{\theta} (1 + \sin^2\theta) + 4m l^2 \dot{\theta}^2 \sin\theta \cos\theta - m l^2 \Omega^2 \sin\theta \cos\theta + 3mgl \sin\theta &= 0 \end{aligned}$$

At equilibrium, we have: $\dot{\theta} = \ddot{\theta} = 0$ so the Lagrange equation becomes:

$$m l \sin\theta (l \Omega^2 \cos\theta - 3g) = 0$$

so there are 3 solutions for θ @ equilibrium

$$\begin{cases} \theta_1 = 0 \\ \theta_2 = \pi \\ \theta_3 = \arccos \frac{3g}{l \Omega^2} \end{cases}$$

Obviously, the third solution only exists for $\frac{3g}{l \Omega^2} \leq 1$,
 and, for fast rotation $\Omega \rightarrow \infty$ becomes $\theta_3 = \pi/2$ ($= \arccos 0$)
 For slow rotation, there are only two equilibrium solutions θ_1 & θ_2 (as $\frac{3g}{l \Omega^2} > 1$ if Ω is small).